

Problem-Solution Fit Canvas

Date	25-06-2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) - Students (School & College) - Self-Learners - Educators & Teachers - Online Learning Platform Users - Academic Mentors	6. CUSTOMER LIMITATIONS ☐ Limited study time. ☐ Low-end devices. ☐ Internet connectivity issues. ☐ Lack of personalized tutoring. ☐ Difficulty expressing learning problems clearly.	10. YOUR SOLUTION The Emotion Detection & Learning Support Engine is an AI-powered platform that detects learners' emotions from text using BiLSTM and BERT deep learning models. Based on the detected emotion, Gemini AI generates personalized learning guidance, motivational feedback, and study recommendations. The application provides an interactive Streamlit interface, stores emotion records and analytics through CSV logging, and helps students overcome learning challenges while improving engagement and academic performance.
2. PROBLEMS / PAINS + ITS FREQUENCY - Students struggle to understand difficult concepts. - Lack of personalized academic guidance. - Learning frustration and stress during exams. - Difficulty identifying emotional state while studying. - Frequent loss of motivation and confidence.	9. PROBLEM ROOT / CAUSE [RC] ☐ Lack of personalized academic support. ☐ No system to recognize learner emotions. ☐ Generic learning resources for every student. ☐ Difficulty identifying confusion and frustration. ☐ Limited one-to-one mentoring.	7. BEHAVIOR + ITS INTENSITY Behavior - Searches repeatedly for explanations. - Watches multiple videos. - Switches between different resources. - Asks friends or teachers for help. - Gives up when unable to understand. Intensity High, especially during assignments and examinations.

3. TRIGGERS TO ACT

- Poor academic performance.
- Confusing study materials.
- Upcoming examinations.
- Feeling frustrated while learning.
- Need for immediate academic assistance.

4. EMOTIONS BEFORE / AFTER [EM]

Before

- 😞 Confused
- 😡 Frustrated
- 😴 Bored
- 😰 Anxious
- 😞 Unmotivated

After

- 😊 Confident
- 😊 Motivated
- 📖 Focused
- 💡 Curious
- 🎯 Ready to Learn

5. AVAILABLE SOLUTIONS (Pros & Cons)

Existing Solutions

- Google Search
- YouTube Tutorials
- Chatbots
- Online Learning Platforms

Pros

- Easy access to learning resources.
- Large amount of educational content.
- Available anytime.

Cons

- No emotion detection.
- No personalized learning support.
- Generic responses.
- Cannot understand learner emotions.

8. CHANNELS OF BEHAVIOR

Online

- Streamlit Web Application
- Google Search
- YouTube
- Educational Websites
- Learning Management Systems (LMS)

Offline

- Classroom Learning
- Teachers
- Friends
- Study Groups
- Printed Notes & Books